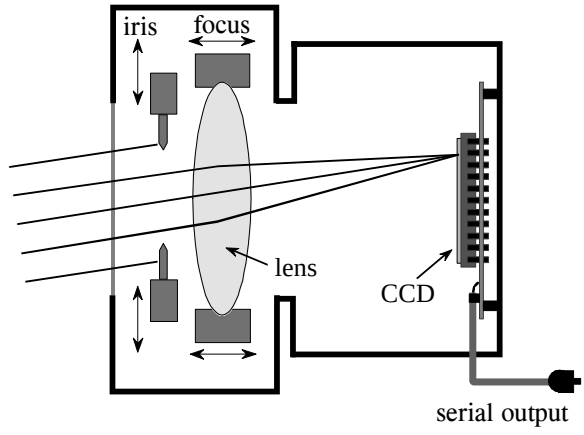

FIGURE 23-3
Diagram of an electronic camera. Focusing is achieved by moving the lens toward or away from the imaging sensor. The amount of light reaching the sensor is controlled by the iris, a mechanical device that changes the effective diameter of the lens. The most common imaging sensor in present day cameras is the CCD, a two-dimensional array of light sensitive elements.

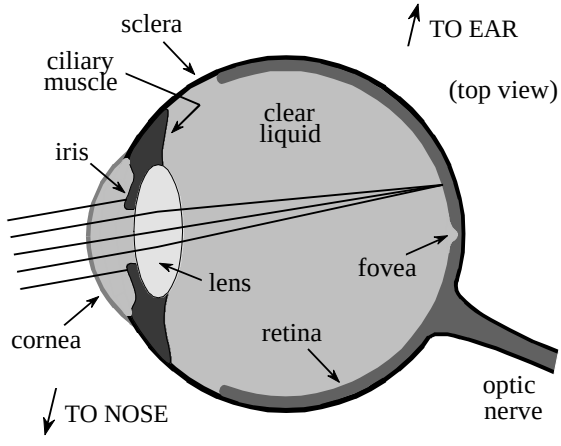

FIGURE 23-4
Diagram of the human eye. The eye is a liquid filled sphere about 3 cm in diameter, enclosed by a tough outer case called the sclera. Focusing is mainly provided by the cornea, a fixed lens on the front of the eye. The focus is adjusted by contracting muscles attached to a flexible lens within the eye. The amount of light entering the eye is controlled by the iris, formed from opaque muscle tissue covering a portion of the lens. The rear hemisphere of the eye contains the retina, a layer of light sensitive nerve cells that converts the image to a neural signal in the optic nerve.

but can degrade the final image by increasing its dynamic range. DSP can improve this situation by suppressing the illumination signal, allowing the reflectance signal to dominate the image. The next chapter presents an approach for implementing this algorithm.

The light sensitive surface that covers the rear of the eye is called the **retina**. As shown in Fig. 23-5, the retina can be divided into three main layers of specialized nerve cells: one for converting light into neural signals, one for image processing, and one for transferring information to the optic nerve leading to the brain. In nearly all animals, these layers are seemingly *backward*. That is, the light sensitive cells are in last layer, requiring light to pass through the other layers before being detected.

There are two types of cells that detect light: **rods** and **cones**, named for their physical appearance under the microscope. The rods are specialized in operating with very little light, such as under the nighttime sky. Vision appears very *noisy* in near darkness, that is, the image appears to be filled with a continually changing grainy pattern. This results from the image signal being very weak, and is not a limitation of the eye. There is so little light entering